

1. Administrative & Study Identification

Full Study Title A Randomized, Multicenter, Open-Label, Phase III Clinical Trial to Evaluate the Efficacy, Safety, and Pharmacokinetics of Prophylactic Emicizumab Versus No Prophylaxis in Hemophilia A Patients **Brief Title** Efficacy, Safety, and Pharmacokinetic Study of Prophylactic Emicizumab Versus No Prophylaxis in Hemophilia A Participants **Short Title/Acronym** HAVEN 5 **Protocol Number** Y039309 **NCT Number** NCT03315455 **Sponsor Name** Roche / Hoffmann-La Roche **Investigational Product** Emicizumab (Trade Name: Hemlibra) **Phase of Development** Phase III **Trial Status** Completed **Medical Monitor** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of prophylactic emicizumab versus no prophylaxis in preventing bleeds (Annualized Bleeding Rate [ABR] for treated bleeds) in participants with hemophilia A. **Secondary Objectives:** To evaluate the ABR for all bleeds, treated target joint bleeds, overall safety (including adverse events and thromboembolic events), and the pharmacokinetics of emicizumab.

2.2 Background & Rationale To address the need for effective, lower-burden prophylactic treatments in patients with severe hemophilia A without factor VIII (FVIII) inhibitors, or hemophilia A of any severity with FVIII inhibitors. This study investigates whether subcutaneously administered emicizumab (at different dosing intervals, weekly vs. every 4 weeks) can effectively control bleeding and improve the use of prophylaxis across the Asia-Pacific region.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** Completed **Control Type:** Active Comparator / No prophylaxis (Control arm) **Blinding:** Open-label **Randomization:** 2:2:1 ratio for randomized arms (Arm A: Emicizumab QW, Arm B: Emicizumab Q4W, Arm C: No prophylaxis). Arm D consists of a non-randomized pediatric cohort.

3.2 Planned Enrollment & Duration Target **SN\$:** 85 participants **Number of Sites:** Multicenter (Asia-Pacific region) **Estimated Study Start:** April 2018 **Estimated Primary Completion:** January 2019 (Enrollment completion / primary data cutoff)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria For Arms A, B, and C: 1. Diagnosis of severe congenital hemophilia A (without FVIII inhibitors) or hemophilia A with FVIII inhibitors. 2. Aged ≥ 12 years at the time of informed consent. 3. Body weight ≥ 40 kg at the time of screening. 4. Participants without FVIII inhibitors (<0.6 BU/mL) who completed successful immune tolerance induction (ITI) must have done so at least 5 years before screening and have no evidence of inhibitor recurrence. 5. Experienced at least 5 bleeds over the prior 24 weeks while receiving episodic therapy with FVIII or bypassing agents, with documented details. 6. Adequate hematologic, hepatic, and renal function. 7. For women of childbearing potential: agreement to remain abstinent or use protocol-defined contraception during the treatment period and for at least 5 elimination half-lives (24 weeks) after the last dose.

For Arm D: 1. Diagnosis of congenital hemophilia A of any severity and documented history of high-titer inhibitor (≥ 5 BU/mL). 2. Children <12 years old at time of informed consent. 3. Body weight >3 kg at time of informed consent. 4. Requires treatment with bypassing agents. 5. Adequate hematologic, hepatic, and renal function. 6. For female participants of childbearing potential, same contraception criteria as Arms A, B, and C.

4.2 Exclusion Criteria For Arms A, B, and C: 1. Inherited or acquired bleeding disorder other than hemophilia A. 2. High risk for thrombotic microangiopathy (TMA) or previous (past 12 months)/current treatment for thromboembolic disease. 3. History of illicit drug or alcohol abuse within 48 weeks prior to screening. 4. Known human immunodeficiency virus (HIV) infection with CD4 count <200 cells/mL within 24 weeks prior to screening. 5. Use of systemic immunomodulators at enrollment or planned use during the study (except anti-retroviral therapy). 6. Planned surgery (excluding minor procedures such as tooth extraction) during the study. 7. Receipt of emicizumab in a prior investigational study, or concurrent use of a non-hemophilia-related investigational drug. 8. Pregnant or lactating, or intending to become pregnant.

For Arm D: 1. Inherited or acquired bleeding disorder other than hemophilia A. 2. Ongoing (or planned) ITI therapy or prophylaxis treatment with FVIII. 3. Known infection with HIV, hepatitis B virus (HBV), or hepatitis C virus (HCV). 4. High risk for TMA, thromboembolic disease, or other conditions increasing bleeding/thrombosis risk. 5. Use of systemic immunomodulators at enrollment or planned use during the study. 6. Inability (or unwillingness by caregiver) to receive blood or blood products. 7. Planned surgery (excluding minor procedures) during the study. 8. Receipt of emicizumab in a prior investigational study, or

concurrent use of a non-hemophilia-related investigational drug.
9. Pregnant or lactating, or intending to become pregnant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A:** Emicizumab 3 mg/kg once every week (QW) for 4 weeks, followed by 1.5 mg/kg QW. * **Arm B:** Emicizumab 3 mg/kg QW for 4 weeks, followed by 6 mg/kg once every 4 weeks (Q4W). * **Arm C:** No prophylaxis (Control). * **Arm D:** Emicizumab prophylaxis at 3 mg/kg QW for 4 weeks, followed by 1.5 mg/kg QW. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** At least 24 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care treatments on an episodic basis for the treatment of breakthrough bleeds. **Prohibited:** Other investigational drugs or bypassing agents used prophylactically outside the study parameters. Ongoing ITI or FVIII prophylaxis (specifically for Arm D).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Bleed Rate evaluation, FVIII inhibitor testing
Baseline	Day 0	Randomization, First Dose (Loading Dose 3 mg/kg), Bleed Diary training
Interim	Weeks 4, 8, 12, 16, 20	Safety Labs, Efficacy (Bleed) Assessment, PK sampling, Dose transition (at Week 4)
End of Study	Week 24+	Final Vital Signs, Primary Endpoint Assessment, Option to transition to extension study/marketed drug

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes
1. **Model-Based Annualized Bleeding Rate for Treated Bleeds:** Using a negative binomial regression model to account for different follow-up times (72-hour rule implemented; surgery/procedure bleeds excluded).
2. **Mean Calculated Annualized Bleeding Rate for Treated Bleeds:** (number of bleeds/number of days) x 365.25.
3. **Median Calculated Annualized Bleeding Rate for Treated Bleeds.**

7.2 Secondary & Exploratory Outcomes
1. **Model-Based Annualized Bleeding Rate for All Bleeds.**
2. **Mean Calculated Annualized Bleeding Rate for All Bleeds.**
3. **Median Calculated Annualized Bleeding Rate for All Bleeds.**
4. Annualized bleeding rate (ABR) for treated target joint bleeds.
5. Incidence of Adverse Events (AEs), Thrombotic Microangiopathies (TMAs), and Thromboembolic Events.
6. Pharmacokinetics of Emicizumab.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient diaries and HCP interviews at every visit (monitoring for events like upper respiratory tract infections, injection site reactions). **Serious Adverse Events (SAEs):** Specific 24-hour reporting protocols for severe events such as Thrombotic Microangiopathy (TMA) or thromboembolic events. **Data Monitoring Committee (DMC):** External independent committee to evaluate ongoing safety and risk of thrombosis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for the primary ABR efficacy evaluation. Safety Set for AE/SAE and ADA (Anti-Drug Antibody) evaluations. **Sample Size Justification:** Target of ~85 patients powered to demonstrate a statistically significant reduction in ABR. **Handling of Missing Data:** Negative binomial regression models adapted to account for differing lengths of observation periods and missing diary inputs.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite visits to verify eCRF against source documents (bleed diaries, FVIII administration logs). **Audit Trail:** eCRF locking, tracking of adherence to SC injection schedules, and stringent tracking of breakthrough bleed treatments.

11. Study Identifiers & Governance

Primary ID: Y039309 **Registry Number:** NCT03315455 **Sponsor/Lead Organization:** Roche / Hoffmann-La Roche **Study Title:** HAVEN 5 **Official Regulatory Title:** A Randomized, Multicenter, Open-Label, Phase III Clinical Trial to Evaluate the Efficacy, Safety, and Pharmacokinetics of Prophylactic Emicizumab Versus No Prophylaxis in Hemophilia A Patients

12. Clinical Framework (Hemophilia Specific)

Primary Condition: Hemophilia A **Diagnosis Threshold:** Severe congenital hemophilia A without inhibitors or hemophilia A with inhibitors (including a pediatric cohort with high-titer inhibitors). **Medical Methodology:** Subcutaneous Monoclonal Antibody Prophylaxis **Optimization Target:** Significant reduction in Annualized Bleeding Rate (ABR) and target joint bleeds

13. Study Procedures & Methodology

Management Pathway: Emicizumab (Hemlibra) subcutaneous prophylactic pathway (QW or Q4W dosing) **Education Protocol (HCP):** Injection administration, monitoring for TMA and thromboembolic events **Education Protocol (Patient):** SC self-injection training, proper logging of treated bleeds into electronic diaries **Quality Audit Mechanism:** Bleed diary compliance tracking and injection adherence monitoring

14. Observation & Follow-Up Schedule

Total Duration: At least 24 weeks (primary evaluation), followed by an optional extension **Onsite Visit Cadence:** Weeks 4, 8, 12, 16, 20, 24 **Remote Monitoring Frequency:** Continuous via electronic bleed tracking diaries **Checklist Review Interval:** Every 4 weeks (coinciding with Q4W administration cycles)

15. Sign-off & Approval

Role	Name	Signature	Date		:--		:--		:--		:--	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]									
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]									
Statistician	[Name]	_____	[DD-MMM-YYYY]									